Documentation of calibration measurements

1. First, I used AI to analyze the script's history and ensure there were no large functional changes (besides bug fixes) that could break the execution of the program compared to before. I will note, the bug fix on 2026-06-02, which I reviewed manually, was a valid bug and could have lead to weird calibration performance before

- Change history

Date	Commit	Effect
2026-06-02	`54ecaed2`	Fixed randomized-setting bookkeeping. Filenames and completion flags now use the original scalar index rather than its shuffled position.
2026-06-02	`e6cfd773`	Corrected the displayed setting progress index.
2026-06-16	`bfda0fbc`	Added the persistent Bluetooth client, structured error capture, and failure diagnostics.
2026-06-17	`1833b04c`	Added five retries for transient Bluetooth state-read failures.
2026-06-22	`5fd79f59`	Added the inter-NDF CombiLED cooldown callback.
2026-06-24	`f338102a`	Switched to `initialize_combiLED_lightLogger(...)` and rounded the NDF passed to it. This is the most recent functional change in the file.
2026-07-01	`6ca15894`	Documentation expansion only; collection behavior was unchanged.

1. Then, I gave the previously saved calibration file (metadata included) to AI to ensure the code uses the exact same parameters used to collect the previous one

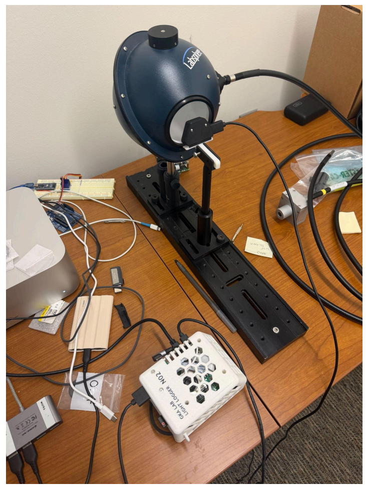
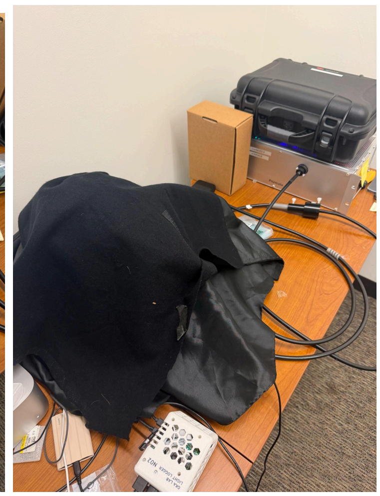
- Yes—the current code still uses the same core MS-linearity acquisition settings found in MS_linearity_data.mat.

Setting	Saved file	Current code
CombiLED background	`[1 1 1 1 1 1 1 1]`	Same
Intensity scalars	`0.05:0.10:0.95`	Same
Number of intensity levels	10	Same
Repetitions per level	3	Same
Recording duration	12 seconds	Same
Minispect indicator LED	Off	Same
Completion dimensions	`NDF × 10 × 3`	Same
Saved NDFs	`0:4`	Caller-controlled

These current defaults are defined at collect_light_logger_calibration_data.m (line 401).

1. I took every NDF 0-3 and cleaned the lens with lens cleaner. I inspected it for damage to the glass. I found NDF 3.0 has a spattering of artifacts on the glass that do not come off with the lens cleaner. They look more surface level than deep into the glass

2. I removed the IR filter box from the optical pathway between the sphere and the CombiLED
3. I powered on device Number 2 (the one with 02 written on the shell. At the time of writing, both 01 and 02 share the same “Device 2” bluetooth address, this should be changed at some point) with the black camera (the second camera sensor block)
4. I configured the device and camera with respect to the sphere and shrouded it with 2 layers of black cloth, deliberately not touching the sphere as I did so. The camera and the sphere are both screwed into their locations.

5. I turned on the CombiLED and inserted 0 NDF into the cassette
6. I ejected the external SSD from my Mac Studio and plugged it into the Raspberry Pi
7. I turned off all the lights in the room and dimmed my monitor to the lowest brightness. I attempted to reduce the effect of the LED lights on the CombiLED by placing a box on top of it
8. I tbUseProject’ed and ran the following code
    - experiment_name = "MSLinearityMeasurements";
    - device_num = 2;
    - sensor_ids = [2, 2, 2];
    - NDFs.ms_linearity = [0, 1, 2, 3];
    - NDFs.world_linearity = [];
    - NDFs.temporal_sensitivity = [];
    - NDFs.phase_fitting = [];
    - NDFs.contrast_gamma = [];
    - calibration_metadata = false;
    - upload_video_data = false;
    - collect_light_logger_calibration_data(experiment_name, device_num, sensor_ids, NDFs, calibration_metadata, upload_video_data)
9. After doing this, I got an error that the CombiLED Arduino could not be found. I changed the USB-C port it was plugged into and tried again, changing the experiment_name to “MSLinearityMeasurements2”
10. I watched the execution of the program without changing any windows on my monitor
11. NDF 0 - Executed fine

12. During the 5 minute cooldown period, I turned on the lights and inserted NDF1
13. NDF 1 executed fine
14. During the execution of NDF 2, I heard a click from the direction of the sphere. I did not adjust or touch anything before or after this, but it otherwise executed fine
15. NDF 3 executed fine